ecbench report: ECBS1ha345d9b7f6b2

Label: comprehensive generic sweep, prime-order curves over 16..32-bit primes

Description: Method × size scaling study (prime curves, bits of p 16,18,20,22,24,26,28,30,32; 2 independent single-target workload(s) per curve, 3 interleaved round(s)). One curve per size so `ecbench bound fit` and `ecbench report` can fit cost against r. Counted group operations; a measurement, not a performance claim.

Generated by `ecbench report` (ecbench.report/v1). Observations only; derived numbers carry the key of their derivation (last section).

Session

field	value
session id	ECBS1ha345d9b7f6b2
spec id	ECS1h5fd596b5123e
commit	f793f1fe5edc540668ed836d5e975eb040a34dff (dirty)
status	complete
run options	cpus="auto", lock=/tmp/crypto-bench.lock, wait_for_lock=false, allow_busy=true

Host and isolation

Intel(R) Xeon(R) Processor @ 2.10GHz; 4 logical / 4 physical cores; kernel 6.18.44-fc-v139; 15.7 GiB; env class ECBENV2he1389885ccd0; virtualization cpu_flag=hypervisor, systemd_detect_virt_vm=kvm.

Isolation earned by the 378 measured records: L0 0, L1 91, L2 287, L3 0. Required of a wall-clock figure: L2.

blocker	measured records
L3: governor is None, not performance	378
L3: not known to be bare metal	378
L3: run CPU is in neither isolcpus nor nohz_full	378
L3: turbo/boost is not known to be off	378
L2: 1 ticks of hypervisor steal on the run CPU	22
L2: 129.4 preemptions per second	2
L2: 2 ticks of hypervisor steal on the run CPU	2
L2: 50.6 preemptions per second	2
L2: 52.7 preemptions per second	2
L2: 55.8 preemptions per second	2
L2: 60.0 preemptions per second	2
L2: 60.6 preemptions per second	2

Arms

arm	role	method	params	expected S [d4]
rho-plain	baseline	rho.plain	cap_multiple=64	1.2533
rho-negation	reference	rho.negation	cap_multiple=64	0.8862
bsgs-textbook	candidate	bsgs.textbook	defaults	1.5000
bsgs-interleaved	candidate	bsgs.interleaved	defaults	1.3333
bsgs-negation	candidate	bsgs.negation	defaults	1.0000
kangaroo-vow	candidate	kangaroo.vow	cap_multiple=64	2.0000
rho-negation-aa	control	rho.negation	cap_multiple=64	0.8862

Workloads

workload	curve	family	log2 r	A	target index	target seed
W43a0abf0d808	icv1-fp16-t295-8d3c3165	prime	15.697	2	0	20261002
Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	prime	15.697	2	1	20261002
Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	prime	17.697	2	0	20261002
Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	prime	17.697	2	1	20261002

workload	curve	family	log2 r	A	target index	target seed
W141767324d6e	icv1-fp20-t727-cd198a38	prime	19.101	2	0	20261002
W8d7a772f8ccd	icv1-fp20-t727-cd198a38	prime	19.101	2	1	20261002
W9a300e433cac	icv1-fp22-tm1385-475dcb5f	prime	21.216	2	0	20261002
W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	prime	21.216	2	1	20261002
Wec44ca6ca670	icv1-fp24-t1059-6df599df	prime	23.974	2	0	20261002
W3256bf1f8f74	icv1-fp24-t1059-6df599df	prime	23.974	2	1	20261002
Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	prime	25.346	2	0	20261002
Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	prime	25.346	2	1	20261002
W1062bb0f2e32	icv1-fp28-tm3161-380f655a	prime	27.955	2	0	20261002
W08bb71dc3759	icv1-fp28-tm3161-380f655a	prime	27.955	2	1	20261002
W0d13e64ff5c5	icv1-fp30-t31711-b403411a	prime	29.852	2	0	20261002
W1670e908d864	icv1-fp30-t31711-b403411a	prime	29.852	2	1	20261002
W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	prime	31.691	2	0	20261002
W751e52089a7e	icv1-fp32-t74357-b7396dac	prime	31.691	2	1	20261002

Runs

Planned 504 (126 warm-up, 378 measured); records 504 (378 measured); verified 378 (378 with matches_target); not verified 0; planned without a record 0.

No record has a status other than verified.

Graphs

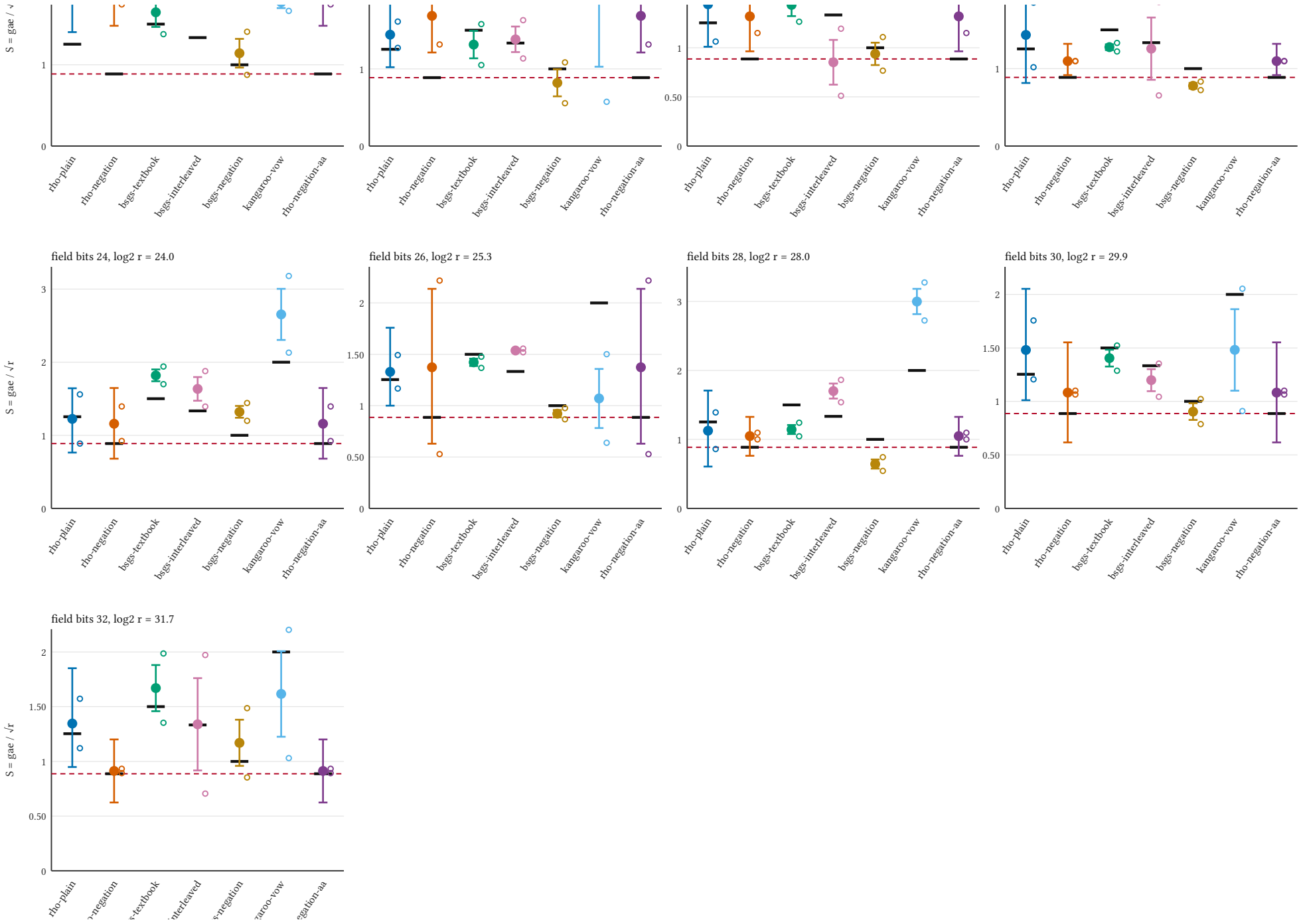


Figure 1: Mean S by arm per curve with 95% bootstrap whiskers, the floor and expected S; session ECBS1ha345d9b7f6b2, n = 378 verified runs, unit gae (S = gae / √r)

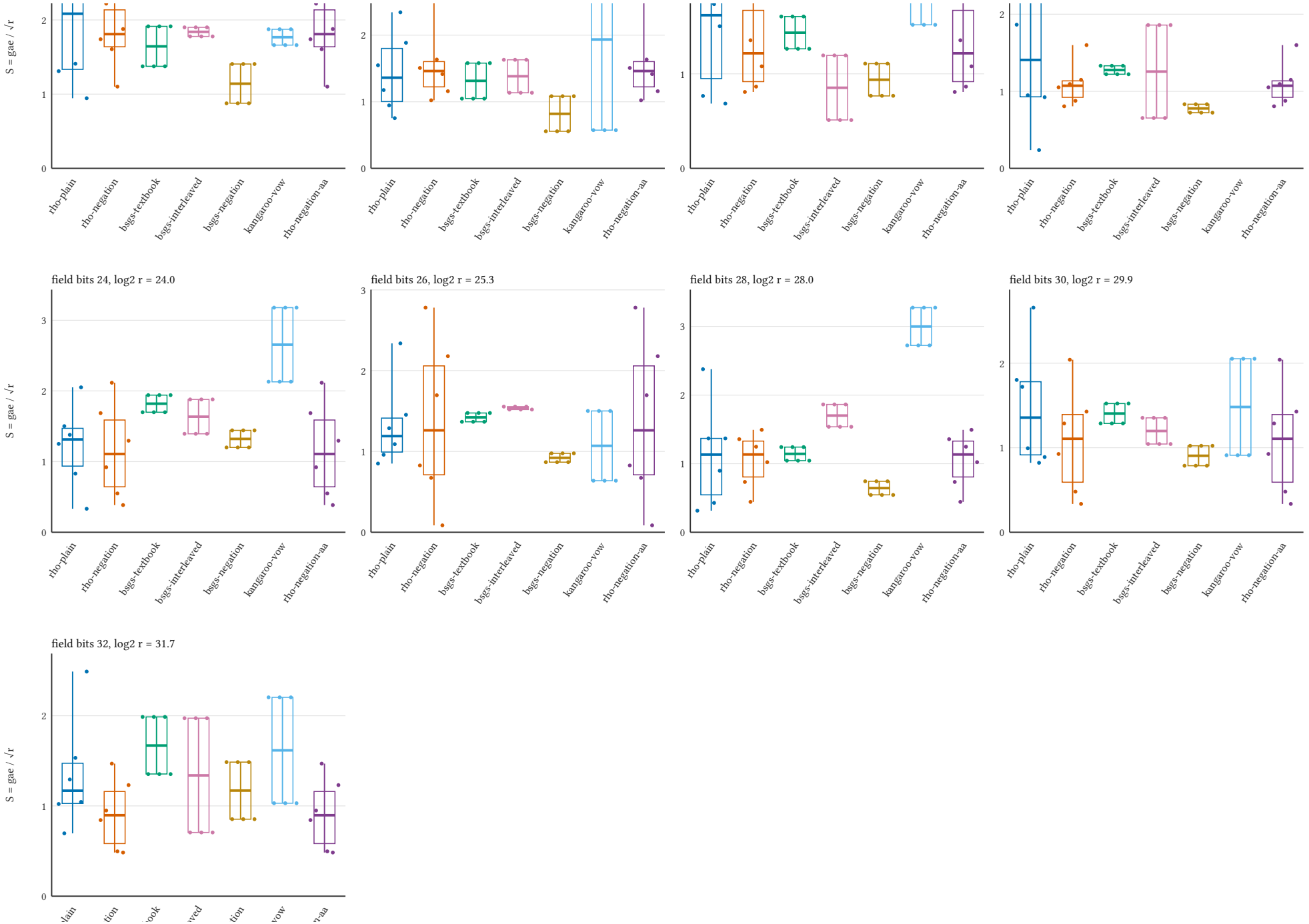
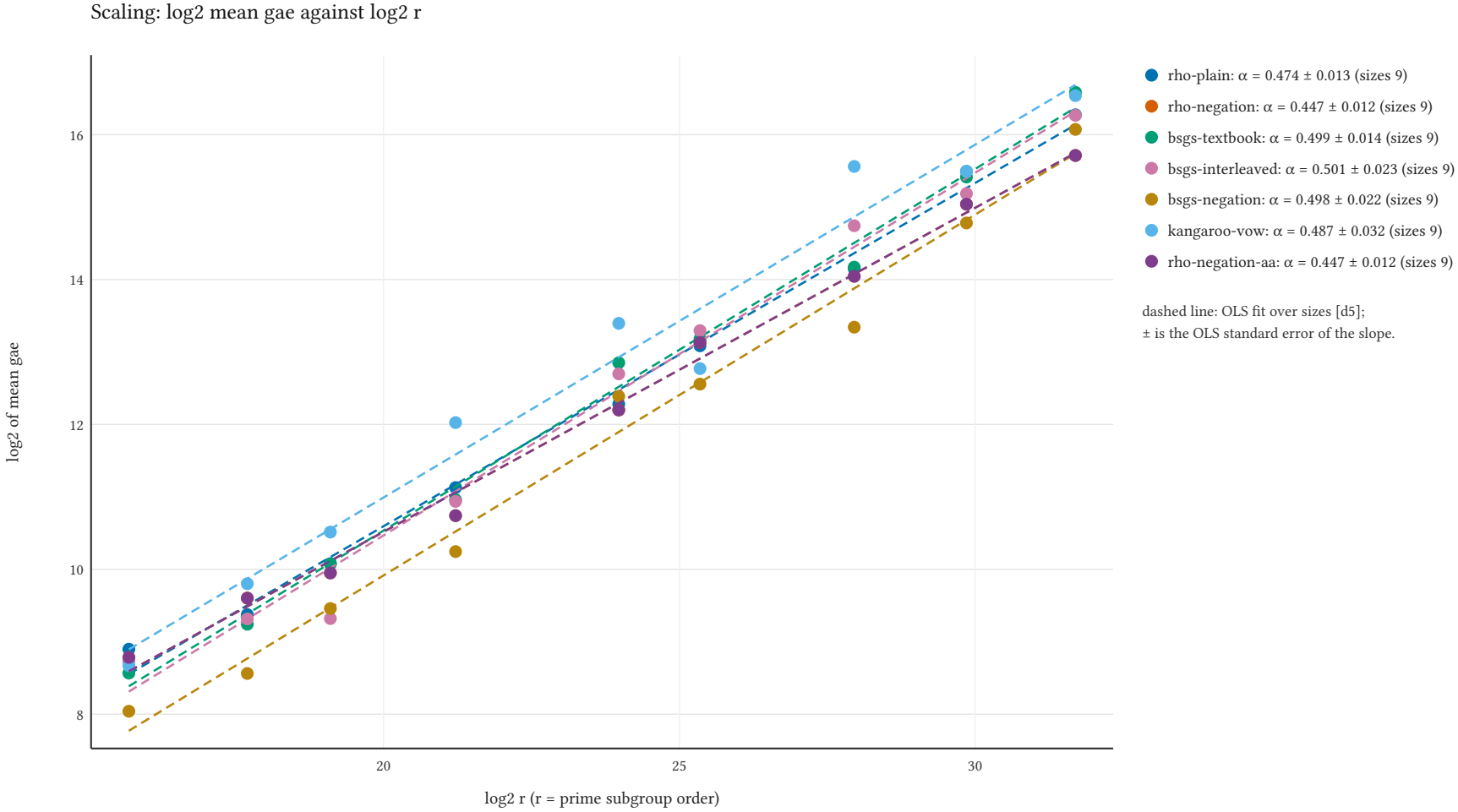


Figure 2: Distribution of S over the verified runs, per arm and curve; session ECBS1ha345d9b7f6b2, n = 378 verified runs, unit gae (S = gae / √r)



Session ECBS1ha345d9b7f6b2, n = 378 verified runs over 9 sizes. Unit: gae (counted group operations).

Figure 3: log2 mean gae against log2 r with OLS fits and fitted exponents; session ECBS1ha345d9b7f6b2, n = 378 verified runs, unit gae (S = gae / √r)

Results per arm and curve

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-plain	icv1-fp16-t295-8d3c3165	15.697	6 / 6	2.0675	2.0849	0.9459	3.0286	0.9429	[1.4015, 2.7010]	0.8862	1.2533
rho-plain	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.4417	1.3614	0.7529	2.3432	0.6019	[1.0201, 1.8814]	0.8862	1.2533
rho-plain	icv1-fp20-t727-cd198a38	19.101	6 / 6	1.4410	1.6201	0.6854	2.1615	0.5937	[1.0103, 1.8583]	0.8862	1.2533
rho-plain	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	1.4338	1.4067	0.2377	2.3265	0.8561	[0.8144, 2.0159]	0.8862	1.2533
rho-plain	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.2236	1.3142	0.3315	2.0508	0.5894	[0.7635, 1.6436]	0.8862	1.2533

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-plain	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.3297	1.1891	0.8509	2.3373	0.5398	[0.9995, 1.7580]	0.8862	1.2533
rho-plain	icv1-fp28-tm3161-380f655a	27.955	6 / 6	1.1255	1.1322	0.3138	2.3770	0.7591	[0.6059, 1.7084]	0.8862	1.2533
rho-plain	icv1-fp30-t31711-b403411a	29.852	6 / 6	1.4806	1.3569	0.8214	2.6571	0.7163	[1.0113, 2.0524]	0.8862	1.2533
rho-plain	icv1-fp32-t74357-b7396dac	31.691	6 / 6	1.3463	1.1692	0.6968	2.4892	0.6265	[0.9480, 1.8517]	0.8862	1.2533
rho-negation	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.9113	1.8093	1.1021	2.9201	0.6153	[1.4789, 2.3879]	0.8862	0.8862
rho-negation	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.6851	1.4612	1.0219	3.3738	0.8574	[1.2121, 2.4036]	0.8862	0.8862
rho-negation	icv1-fp20-t727-cd198a38	19.101	6 / 6	1.3185	1.2167	0.8081	2.0295	0.4981	[0.9630, 1.6992]	0.8862	0.8862
rho-negation	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	1.0958	1.0726	0.8048	1.5993	0.2796	[0.9151, 1.3196]	0.8862	0.8862
rho-negation	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.1585	1.1079	0.3845	2.1159	0.6693	[0.6807, 1.6464]	0.8862	0.8862
rho-negation	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.3732	1.2605	0.0859	2.7800	1.0178	[0.6304, 2.1375]	0.8862	0.8862
rho-negation	icv1-fp28-tm3161-380f655a	27.955	6 / 6	1.0486	1.1339	0.4423	1.4923	0.3997	[0.7623, 1.3275]	0.8862	0.8862
rho-negation	icv1-fp30-t31711-b403411a	29.852	6 / 6	1.0827	1.1059	0.3359	2.0404	0.6361	[0.6166, 1.5520]	0.8862	0.8862
rho-negation	icv1-fp32-t74357-b7396dac	31.691	6 / 6	0.9122	0.8964	0.4845	1.4684	0.3932	[0.6238, 1.2003]	0.8862	0.8862
bsgs-textbook	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.6445	1.6445	1.3754	1.9135	0.2947	[1.4651, 1.8238]	0.8862	1.5000
bsgs-textbook	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.3137	1.3137	1.0479	1.5795	0.2911	[1.1365, 1.4909]	0.8862	1.5000
bsgs-textbook	icv1-fp20-t727-cd198a38	19.101	6 / 6	1.4348	1.4348	1.2641	1.6054	0.1870	[1.3210, 1.5485]	0.8862	1.5000
bsgs-textbook	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	1.2760	1.2760	1.2212	1.3308	0.0600	[1.2395, 1.3125]	0.8862	1.5000
bsgs-textbook	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.8198	1.8198	1.6983	1.9412	0.1330	[1.7388, 1.9007]	0.8862	1.5000
bsgs-textbook	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.4217	1.4217	1.3675	1.4759	0.0594	[1.3855, 1.4578]	0.8862	1.5000
bsgs-textbook	icv1-fp28-tm3161-380f655a	27.955	6 / 6	1.1433	1.1433	1.0442	1.2424	0.1086	[1.0772, 1.2093]	0.8862	1.5000
bsgs-textbook	icv1-fp30-t31711-b403411a	29.852	6 / 6	1.4047	1.4047	1.2873	1.5222	0.1287	[1.3264, 1.4830]	0.8862	1.5000
bsgs-textbook	icv1-fp32-t74357-b7396dac	31.691	6 / 6	1.6696	1.6696	1.3532	1.9861	0.3467	[1.4587, 1.8806]	0.8862	1.5000
bsgs-interleaved	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.8397	1.8397	1.7790	1.9005	0.0665	[1.7992, 1.8802]	0.8862	1.3333
bsgs-interleaved	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.3821	1.3821	1.1347	1.6294	0.2709	[1.2172, 1.5469]	0.8862	1.3333
bsgs-interleaved	icv1-fp20-t727-cd198a38	19.101	6 / 6	0.8521	0.8521	0.5107	1.1934	0.3739	[0.6245, 1.0796]	0.8862	1.3333
bsgs-interleaved	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	1.2565	1.2565	0.6535	1.8594	0.6605	[0.8545, 1.6584]	0.8862	1.3333
bsgs-interleaved	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.6353	1.6353	1.3924	1.8782	0.2660	[1.4734, 1.7972]	0.8862	1.3333
bsgs-interleaved	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.5373	1.5373	1.5191	1.5555	0.0200	[1.5251, 1.5494]	0.8862	1.3333
bsgs-interleaved	icv1-fp28-tm3161-380f655a	27.955	6 / 6	1.7008	1.7008	1.5380	1.8637	0.1784	[1.5922, 1.8094]	0.8862	1.3333
bsgs-interleaved	icv1-fp30-t31711-b403411a	29.852	6 / 6	1.1984	1.1984	1.0437	1.3530	0.1694	[1.0953, 1.3015]	0.8862	1.3333
bsgs-interleaved	icv1-fp32-t74357-b7396dac	31.691	6 / 6	1.3388	1.3388	0.7059	1.9717	0.6933	[0.9169, 1.7607]	0.8862	1.3333
bsgs-negation	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.1411	1.1411	0.8765	1.4058	0.2899	[0.9647, 1.3176]	0.8862	1.0000
bsgs-negation	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	0.8190	0.8190	0.5554	1.0826	0.2888	[0.6433, 0.9948]	0.8862	1.0000
bsgs-negation	icv1-fp20-t727-cd198a38	19.101	6 / 6	0.9374	0.9374	0.7667	1.1081	0.1870	[0.8236, 1.0512]	0.8862	1.0000
bsgs-negation	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	0.7775	0.7775	0.7227	0.8323	0.0600	[0.7410, 0.8141]	0.8862	1.0000
bsgs-negation	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.3201	1.3201	1.1988	1.4414	0.1329	[1.2393, 1.4010]	0.8862	1.0000
bsgs-negation	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	0.9219	0.9219	0.8677	0.9761	0.0594	[0.8858, 0.9580]	0.8862	1.0000
bsgs-negation	icv1-fp28-tm3161-380f655a	27.955	6 / 6	0.6434	0.6434	0.5443	0.7424	0.1085	[0.5773, 0.7094]	0.8862	1.0000
bsgs-negation	icv1-fp30-t31711-b403411a	29.852	6 / 6	0.9048	0.9048	0.7873	1.0222	0.1287	[0.8265, 0.9831]	0.8862	1.0000
bsgs-negation	icv1-fp32-t74357-b7396dac	31.691	6 / 6	1.1695	1.1695	0.8531	1.4860	0.3466	[0.9585, 1.3805]	0.8862	1.0000
kangaroo-vow	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.7681	1.7681	1.6618	1.8744	0.1165	[1.6972, 1.8390]	0.8862	2.0000
kangaroo-vow	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.9353	1.9353	0.5728	3.2978	1.4926	[1.0270, 2.8437]	0.8862	2.0000
kangaroo-vow	icv1-fp20-t727-cd198a38	19.101	6 / 6	1.9475	1.9475	1.5188	2.3762	0.4696	[1.6617, 2.2333]	0.8862	2.0000
kangaroo-vow	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	2.6712	2.6712	2.4265	2.9160	0.2681	[2.5081, 2.8344]	0.8862	2.0000
kangaroo-vow	icv1-fp24-t1059-6df599df	23.974	6 / 6	2.6540	2.6540	2.1296	3.1785	0.5745	[2.3044, 3.0037]	0.8862	2.0000
kangaroo-vow	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.0700	1.0700	0.6386	1.5013	0.4725	[0.7824, 1.3575]	0.8862	2.0000

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
kangaroo-vow	icv1-fp28-tm3161-380f655a	27.955	6 / 6	2.9988	2.9988	2.7230	3.2746	0.3021	[2.8150, 3.1827]	0.8862	2.0000
kangaroo-vow	icv1-fp30-t31711-b403411a	29.852	6 / 6	1.4815	1.4815	0.9103	2.0528	0.6258	[1.1007, 1.8623]	0.8862	2.0000
kangaroo-vow	icv1-fp32-t74357-b7396dac	31.691	6 / 6	1.6165	1.6165	1.0300	2.2029	0.6424	[1.2255, 2.0074]	0.8862	2.0000
rho-negation-aa	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.9113	1.8093	1.1021	2.9201	0.6153	[1.4789, 2.3879]	0.8862	0.8862
rho-negation-aa	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.6851	1.4612	1.0219	3.3738	0.8574	[1.2121, 2.4036]	0.8862	0.8862
rho-negation-aa	icv1-fp20-t727-cd198a38	19.101	6 / 6	1.3185	1.2167	0.8081	2.0295	0.4981	[0.9630, 1.6992]	0.8862	0.8862
rho-negation-aa	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	1.0958	1.0726	0.8048	1.5993	0.2796	[0.9151, 1.3196]	0.8862	0.8862
rho-negation-aa	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.1585	1.1079	0.3845	2.1159	0.6693	[0.6807, 1.6464]	0.8862	0.8862
rho-negation-aa	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.3732	1.2605	0.0859	2.7800	1.0178	[0.6304, 2.1375]	0.8862	0.8862
rho-negation-aa	icv1-fp28-tm3161-380f655a	27.955	6 / 6	1.0486	1.1339	0.4423	1.4923	0.3997	[0.7623, 1.3275]	0.8862	0.8862
rho-negation-aa	icv1-fp30-t31711-b403411a	29.852	6 / 6	1.0827	1.1059	0.3359	2.0404	0.6361	[0.6166, 1.5520]	0.8862	0.8862
rho-negation-aa	icv1-fp32-t74357-b7396dac	31.691	6 / 6	0.9122	0.8964	0.4845	1.4684	0.3932	[0.6238, 1.2003]	0.8862	0.8862

arm	curve	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-plain	icv1-fp16-t295-8d3c3165	6 / 6	476.5	2.3329	19.5	305.8	1225.5	567.3	476.5	5.1894	38.5600	admitted (≥L2)
rho-plain	icv1-fp18-tm337-d28d5e09	6 / 6	664.5	1.6268	33.3	472.8	1611.0	758.5	664.5	5.5221	35.6560	informative, L1
rho-plain	icv1-fp20-t727-cd198a38	6 / 6	1080.7	1.6260	26.2	890.7	2451.8	1177.5	1080.7	5.8011	34.1560	informative, L1
rho-plain	icv1-fp22-tm1385-475dcb5f	6 / 6	2237.7	1.6178	58.3	2006.7	4778.3	2338.7	2237.7	6.1917	38.1615	admitted (≥L2)
rho-plain	icv1-fp24-t1059-6df599df	6 / 6	4967.7	1.3807	73.3	4523.2	10583.8	5183.8	4967.7	7.4160	34.4795	informative, L1
rho-plain	icv1-fp26-tm1775-7e8fb6df	6 / 6	8684.2	1.5004	124.8	8186.8	18026.8	8903.7	8684.2	7.3536	35.7015	informative, L1
rho-plain	icv1-fp28-tm3161-380f655a	6 / 6	18154.2	1.2699	130.5	17640.2	36975.3	18376.5	18154.2	7.7509	38.6070	admitted (≥L2)
rho-plain	icv1-fp30-t31711-b403411a	6 / 6	46084.3	1.6707	346.3	45351.5	92849.2	46311.2	46084.3	11.5716	40.8730	informative, L1
rho-plain	icv1-fp32-t74357-b7396dac	6 / 6	79265.7	1.5191	314.3	78555.0	159230.3	79498.7	79265.7	17.7825	50.9690	informative, L1
rho-negation	icv1-fp16-t295-8d3c3165	6 / 6	440.5	2.1567	15.0	254.5	1161.0	533.8	440.5	5.2684	41.8260	informative, L1
rho-negation	icv1-fp18-tm337-d28d5e09	6 / 6	776.7	1.9014	33.8	545.3	1845.3	874.0	776.7	5.8044	34.1200	informative, L1
rho-negation	icv1-fp20-t727-cd198a38	6 / 6	988.8	1.4878	19.5	751.0	2275.7	1088.2	988.8	5.5093	33.3345	informative, L1
rho-negation	icv1-fp22-tm1385-475dcb5f	6 / 6	1710.2	1.2364	44.3	1384.0	3749.8	1820.0	1710.2	6.5425	40.9660	informative, L1
rho-negation	icv1-fp24-t1059-6df599df	6 / 6	4703.2	1.3072	60.7	4127.3	10066.3	4923.2	4703.2	6.9169	35.4545	informative, L1
rho-negation	icv1-fp26-tm1775-7e8fb6df	6 / 6	8968.2	1.5495	125.2	8193.8	18624.8	9197.7	8968.2	7.5639	34.6850	admitted (≥L2)
rho-negation	icv1-fp28-tm3161-380f655a	6 / 6	16915.0	1.1833	123.8	15867.0	34558.5	17157.8	16915.0	9.2794	41.0580	informative, L1
rho-negation	icv1-fp30-t31711-b403411a	6 / 6	33698.2	1.2216	248.5	31969.3	68186.3	33961.5	33698.2	11.3049	43.9070	informative, L1
rho-negation	icv1-fp32-t74357-b7396dac	6 / 6	53706.7	1.0293	190.5	51369.7	108294.3	54000.3	53706.7	15.0524	45.0870	informative, L1
bsgs-textbook	icv1-fp16-t295-8d3c3165	6 / 6	379.0	1.8556	–	368.0	780.0	386.0	378.0	5.1349	36.0640	informative, L1
bsgs-textbook	icv1-fp18-tm337-d28d5e09	6 / 6	605.5	1.4824	–	593.5	1236.0	613.5	604.5	5.5677	35.1670	informative, L1
bsgs-textbook	icv1-fp20-t727-cd198a38	6 / 6	1076.0	1.6190	–	1062.0	2180.0	1085.0	1075.0	5.4638	33.3565	informative, L1
bsgs-textbook	icv1-fp22-tm1385-475dcb5f	6 / 6	1991.5	1.4398	–	1978.5	4014.0	2001.5	1990.5	6.5747	42.7745	informative, L1
bsgs-textbook	icv1-fp24-t1059-6df599df	6 / 6	7388.0	2.0534	–	7370.0	14810.0	7399.0	7387.0	5.8707	33.9125	admitted (≥L2)
bsgs-textbook	icv1-fp26-tm1775-7e8fb6df	6 / 6	9285.0	1.6042	–	9269.0	18607.0	9297.0	9284.0	6.6128	36.0780	informative, L1
bsgs-textbook	icv1-fp28-tm3161-380f655a	6 / 6	18441.5	1.2900	–	18422.5	36923.0	18454.5	18440.5	8.5552	41.4990	admitted (≥L2)
bsgs-textbook	icv1-fp30-t31711-b403411a	6 / 6	43723.0	1.5851	–	43702.0	87489.0	43737.0	43722.0	13.0766	42.9905	admitted (≥L2)
bsgs-textbook	icv1-fp32-t74357-b7396dac	6 / 6	98304.5	1.8840	–	98278.5	196655.0	98319.5	98303.5	25.6615	54.9360	informative, L1
bsgs-interleaved	icv1-fp16-t295-8d3c3165	6 / 6	424.0	2.0759	–	412.0	870.0	431.0	423.0	5.6402	41.6780	informative, L1
bsgs-interleaved	icv1-fp18-tm337-d28d5e09	6 / 6	637.0	1.5595	–	624.0	1299.0	645.0	636.0	5.8546	33.9665	admitted (≥L2)
bsgs-interleaved	icv1-fp20-t727-cd198a38	6 / 6	639.0	0.9614	–	624.0	1306.0	648.0	638.0	5.2260	33.9235	informative, L1
bsgs-interleaved	icv1-fp22-tm1385-475dcb5f	6 / 6	1961.0	1.4178	–	1947.0	3953.0	1971.0	1960.0	6.9255	42.7885	informative, L1
bsgs-interleaved	icv1-fp24-t1059-6df599df	6 / 6	6639.0	1.8452	–	6620.0	13312.0	6650.0	6638.0	7.8597	37.1805	informative, L1

arm	curve	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
bsgs-interleaved	icv1-fp26-tm1775-7e8fb6df	6 / 6	10040.0	1.7346	–	10023.0	20117.0	10052.0	10039.0	7.6955	34.9950	informative, L1
bsgs-interleaved	icv1-fp28-tm3161-380f655a	6 / 6	27435.0	1.9192	–	27415.0	54910.0	27448.0	27434.0	12.9293	42.6515	informative, L1
bsgs-interleaved	icv1-fp30-t31711-b403411a	6 / 6	37300.0	1.3522	–	37278.0	74643.0	37314.0	37299.0	14.0900	46.4680	informative, L1
bsgs-interleaved	icv1-fp32-t74357-b7396dac	6 / 6	78826.0	1.5107	–	78799.0	157698.0	78841.0	78825.0	26.6069	65.9200	informative, L1
bsgs-negation	icv1-fp16-t295-8d3c3165	6 / 6	263.0	1.2876	–	253.0	548.0	270.0	262.0	5.6038	37.4315	informative, L1
bsgs-negation	icv1-fp18-tm337-d28d5e09	6 / 6	377.5	0.9242	–	364.5	780.0	385.5	376.5	5.1167	33.6600	informative, L1
bsgs-negation	icv1-fp20-t727-cd198a38	6 / 6	703.0	1.0577	–	688.0	1434.0	712.0	702.0	4.8961	32.2720	informative, L1
bsgs-negation	icv1-fp22-tm1385-475dcb5f	6 / 6	1213.5	0.8774	–	1199.5	2458.0	1223.5	1212.5	5.2938	37.8985	admitted (≥L2)
bsgs-negation	icv1-fp24-t1059-6df599df	6 / 6	5359.5	1.4896	–	5340.5	10753.0	5370.5	5358.5	5.8472	34.0215	informative, L1
bsgs-negation	icv1-fp26-tm1775-7e8fb6df	6 / 6	6021.0	1.0403	–	6005.0	12079.0	6033.0	6020.0	5.9806	33.5940	informative, L1
bsgs-negation	icv1-fp28-tm3161-380f655a	6 / 6	10378.0	0.7260	–	10359.0	20796.0	10391.0	10377.0	7.2181	38.6675	informative, L1
bsgs-negation	icv1-fp30-t31711-b403411a	6 / 6	28162.0	1.0209	–	28140.0	56367.0	28176.0	28161.0	10.2515	40.5225	informative, L1
bsgs-negation	icv1-fp32-t74357-b7396dac	6 / 6	68858.5	1.3197	–	68839.5	137763.0	68873.5	68857.5	18.6051	52.8145	admitted (≥L2)
kangaroo-vow	icv1-fp16-t295-8d3c3165	6 / 6	407.5	1.9951	51.0	373.5	882.0	429.5	406.5	5.2735	35.3720	informative, L1
kangaroo-vow	icv1-fp18-tm337-d28d5e09	6 / 6	892.0	2.1838	112.0	855.0	1860.0	917.0	891.0	5.3857	32.8590	informative, L1
kangaroo-vow	icv1-fp20-t727-cd198a38	6 / 6	1460.5	2.1975	87.5	1421.5	3006.0	1488.5	1459.5	5.1325	32.6945	informative, L1
kangaroo-vow	icv1-fp22-tm1385-475dcb5f	6 / 6	4169.0	3.0142	247.5	4128.0	8432.0	4200.0	4168.0	6.7939	40.5975	informative, L1
kangaroo-vow	icv1-fp24-t1059-6df599df	6 / 6	10775.0	2.9948	327.5	10722.0	21653.0	10809.0	10774.0	7.0706	34.3915	informative, L1
kangaroo-vow	icv1-fp26-tm1775-7e8fb6df	6 / 6	6988.0	1.2073	219.5	6937.0	14088.0	7025.0	6987.0	6.5426	34.4530	admitted (≥L2)
kangaroo-vow	icv1-fp28-tm3161-380f655a	6 / 6	48372.5	3.3838	770.5	48314.5	96869.0	48413.5	48371.5	11.4399	42.6070	admitted (≥L2)
kangaroo-vow	icv1-fp30-t31711-b403411a	6 / 6	46112.5	1.6717	706.5	46050.5	92358.0	46156.5	46111.5	11.1949	40.5405	informative, L1
kangaroo-vow	icv1-fp32-t74357-b7396dac	6 / 6	95174.0	1.8240	742.0	95113.0	190490.0	95221.0	95173.0	17.1853	46.8540	informative, L1
rho-negation-aa	icv1-fp16-t295-8d3c3165	6 / 6	440.5	2.1567	15.0	254.5	1161.0	533.8	440.5	5.2948	42.8075	informative, L1
rho-negation-aa	icv1-fp18-tm337-d28d5e09	6 / 6	776.7	1.9014	33.8	545.3	1845.3	874.0	776.7	5.5174	32.7785	informative, L1
rho-negation-aa	icv1-fp20-t727-cd198a38	6 / 6	988.8	1.4878	19.5	751.0	2275.7	1088.2	988.8	5.5615	34.4815	admitted (≥L2)
rho-negation-aa	icv1-fp22-tm1385-475dcb5f	6 / 6	1710.2	1.2364	44.3	1384.0	3749.8	1820.0	1710.2	6.7953	40.3125	informative, L1
rho-negation-aa	icv1-fp24-t1059-6df599df	6 / 6	4703.2	1.3072	60.7	4127.3	10066.3	4923.2	4703.2	6.9700	35.2270	informative, L1
rho-negation-aa	icv1-fp26-tm1775-7e8fb6df	6 / 6	8968.2	1.5495	125.2	8193.8	18624.8	9197.7	8968.2	6.9218	35.9040	admitted (≥L2)
rho-negation-aa	icv1-fp28-tm3161-380f655a	6 / 6	16915.0	1.1833	123.8	15867.0	34558.5	17157.8	16915.0	9.1160	40.0520	informative, L1
rho-negation-aa	icv1-fp30-t31711-b403411a	6 / 6	33698.2	1.2216	248.5	31969.3	68186.3	33961.5	33698.2	10.4141	47.0090	informative, L1
rho-negation-aa	icv1-fp32-t74357-b7396dac	6 / 6	53706.7	1.0293	190.5	51369.7	108294.3	54000.3	53706.7	15.0026	47.2210	informative, L1

Results per arm and workload

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-plain	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	2.3401	2.7596	1.3104	2.9505	0.8969	[1.3104, 2.9505]	0.8862	1.2533
rho-plain	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	1.7949	1.4102	0.9459	3.0286	1.0933	[0.9459, 3.0286]	0.8862	1.2533
rho-plain	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	1.6120	1.5469	0.9460	2.3432	0.7009	[0.9460, 2.3432]	0.8862	1.2533
rho-plain	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	1.2714	1.1759	0.7529	1.8854	0.5723	[0.7529, 1.8854]	0.8862	1.2533
rho-plain	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.8192	1.7934	1.5028	2.1615	0.3301	[1.5028, 2.1615]	0.8862	1.2533
rho-plain	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	1.0627	0.7654	0.6854	1.7374	0.5857	[0.6854, 1.7374]	0.8862	1.2533
rho-plain	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0171	0.9483	0.2377	1.8652	0.8159	[0.2377, 1.8652]	0.8862	1.2533
rho-plain	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	1.8504	2.3022	0.9227	2.3265	0.8036	[0.9227, 2.3265]	0.8862	1.2533
rho-plain	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.5598	1.3781	1.2503	2.0508	0.4301	[1.2503, 2.0508]	0.8862	1.2533
rho-plain	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	0.8875	0.8298	0.3315	1.5010	0.5869	[0.3315, 1.5010]	0.8862	1.2533
rho-plain	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.4923	1.2886	0.8509	2.3373	0.7639	[0.8509, 2.3373]	0.8862	1.2533
rho-plain	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.1671	1.0896	0.9587	1.4531	0.2562	[0.9587, 1.4531]	0.8862	1.2533

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-plain	W1062bb0f2e32	icv1-fp28-tm3161-380f655a	3 / 3	0.8596	0.8964	0.3138	1.3687	0.5284	[0.3138, 1.3687]	0.8862	1.2533
rho-plain	W08bb71dc3759	icv1-fp28-tm3161-380f655a	3 / 3	1.3913	1.3681	0.4289	2.3770	0.9743	[0.4289, 2.3770]	0.8862	1.2533
rho-plain	W0d13e64ff5c5	icv1-fp30-t31711-b403411a	3 / 3	1.2055	0.9935	0.8214	1.8015	0.5233	[0.8214, 1.8015]	0.8862	1.2533
rho-plain	W1670e908d864	icv1-fp30-t31711-b403411a	3 / 3	1.7557	1.7203	0.8898	2.6571	0.8841	[0.8898, 2.6571]	0.8862	1.2533
rho-plain	W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	3 / 3	1.1198	1.0445	1.0210	1.2938	0.1511	[1.0210, 1.2938]	0.8862	1.2533
rho-plain	W751e52089a7e	icv1-fp32-t74357-b7396dac	3 / 3	1.5727	1.5322	0.6968	2.4892	0.8969	[0.6968, 2.4892]	0.8862	1.2533
rho-negation	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	1.7414	1.7399	1.6054	1.8788	0.1367	[1.6054, 1.8788]	0.8862	0.8862
rho-negation	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	2.0812	2.2215	1.1021	2.9201	0.9171	[1.1021, 2.9201]	0.8862	0.8862
rho-negation	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	1.3148	1.4146	1.0219	1.5079	0.2579	[1.0219, 1.5079]	0.8862	0.8862
rho-negation	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	2.0554	1.6337	1.1586	3.3738	1.1662	[1.1586, 3.3738]	0.8862	0.8862
rho-negation	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.4877	1.3548	1.0787	2.0295	0.4891	[1.0787, 2.0295]	0.8862	0.8862
rho-negation	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	1.1494	0.8641	0.8081	1.7761	0.5435	[0.8081, 1.7761]	0.8862	0.8862
rho-negation	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0978	1.0944	1.0508	1.1482	0.0488	[1.0508, 1.1482]	0.8862	0.8862
rho-negation	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0937	0.8772	0.8048	1.5993	0.4393	[0.8048, 1.5993]	0.8862	0.8862
rho-negation	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.3951	1.6850	0.3845	2.1159	0.9014	[0.3845, 2.1159]	0.8862	0.8862
rho-negation	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	0.9218	0.9197	0.5495	1.2961	0.3733	[0.5495, 1.2961]	0.8862	0.8862
rho-negation	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	0.5281	0.6728	0.0859	0.8258	0.3906	[0.0859, 0.8258]	0.8862	0.8862
rho-negation	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	2.2182	2.1793	1.6953	2.7800	0.5434	[1.6953, 2.7800]	0.8862	0.8862
rho-negation	W1062bb0f2e32	icv1-fp28-tm3161-380f655a	3 / 3	1.0970	1.3563	0.4423	1.4923	0.5711	[0.4423, 1.4923]	0.8862	0.8862
rho-negation	W08bb71dc3759	icv1-fp28-tm3161-380f655a	3 / 3	1.0003	1.0215	0.7330	1.2463	0.2573	[0.7330, 1.2463]	0.8862	0.8862
rho-negation	W0d13e64ff5c5	icv1-fp30-t31711-b403411a	3 / 3	1.1006	0.9256	0.3359	2.0404	0.8656	[0.3359, 2.0404]	0.8862	0.8862
rho-negation	W1670e908d864	icv1-fp30-t31711-b403411a	3 / 3	1.0647	1.2862	0.4802	1.4275	0.5111	[0.4802, 1.4275]	0.8862	0.8862
rho-negation	W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	3 / 3	0.9321	0.8435	0.4845	1.4684	0.4979	[0.4845, 1.4684]	0.8862	0.8862
rho-negation	W751e52089a7e	icv1-fp32-t74357-b7396dac	3 / 3	0.8922	0.9494	0.4964	1.2309	0.3706	[0.4964, 1.2309]	0.8862	0.8862
bsgs-textbook	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	1.3754	1.3754	1.3754	1.3754	0.0000	[1.3754, 1.3754]	0.8862	1.5000
bsgs-textbook	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	1.9135	1.9135	1.9135	1.9135	0.0000	[1.9135, 1.9135]	0.8862	1.5000
bsgs-textbook	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	1.0479	1.0479	1.0479	1.0479	0.0000	[1.0479, 1.0479]	0.8862	1.5000
bsgs-textbook	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	1.5795	1.5795	1.5795	1.5795	0.0000	[1.5795, 1.5795]	0.8862	1.5000
bsgs-textbook	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.6054	1.6054	1.6054	1.6054	0.0000	[1.6054, 1.6054]	0.8862	1.5000
bsgs-textbook	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	1.2641	1.2641	1.2641	1.2641	0.0000	[1.2641, 1.2641]	0.8862	1.5000
bsgs-textbook	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	1.3308	1.3308	1.3308	1.3308	0.0000	[1.3308, 1.3308]	0.8862	1.5000
bsgs-textbook	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	1.2212	1.2212	1.2212	1.2212	0.0000	[1.2212, 1.2212]	0.8862	1.5000
bsgs-textbook	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.6983	1.6983	1.6983	1.6983	0.0000	[1.6983, 1.6983]	0.8862	1.5000
bsgs-textbook	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	1.9412	1.9412	1.9412	1.9412	0.0000	[1.9412, 1.9412]	0.8862	1.5000
bsgs-textbook	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.3675	1.3675	1.3675	1.3675	0.0000	[1.3675, 1.3675]	0.8862	1.5000
bsgs-textbook	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.4759	1.4759	1.4759	1.4759	0.0000	[1.4759, 1.4759]	0.8862	1.5000
bsgs-textbook	W1062bb0f2e32	icv1-fp28-tm3161-380f655a	3 / 3	1.2424	1.2424	1.2424	1.2424	0.0000	[1.2424, 1.2424]	0.8862	1.5000
bsgs-textbook	W08bb71dc3759	icv1-fp28-tm3161-380f655a	3 / 3	1.0442	1.0442	1.0442	1.0442	0.0000	[1.0442, 1.0442]	0.8862	1.5000
bsgs-textbook	W0d13e64ff5c5	icv1-fp30-t31711-b403411a	3 / 3	1.2873	1.2873	1.2873	1.2873	0.0000	[1.2873, 1.2873]	0.8862	1.5000
bsgs-textbook	W1670e908d864	icv1-fp30-t31711-b403411a	3 / 3	1.5222	1.5222	1.5222	1.5222	0.0000	[1.5222, 1.5222]	0.8862	1.5000
bsgs-textbook	W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	3 / 3	1.9861	1.9861	1.9861	1.9861	0.0000	[1.9861, 1.9861]	0.8862	1.5000
bsgs-textbook	W751e52089a7e	icv1-fp32-t74357-b7396dac	3 / 3	1.3532	1.3532	1.3532	1.3532	0.0000	[1.3532, 1.3532]	0.8862	1.5000
bsgs-interleaved	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	1.9005	1.9005	1.9005	1.9005	0.0000	[1.9005, 1.9005]	0.8862	1.3333
bsgs-interleaved	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	1.7790	1.7790	1.7790	1.7790	0.0000	[1.7790, 1.7790]	0.8862	1.3333
bsgs-interleaved	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	1.6294	1.6294	1.6294	1.6294	0.0000	[1.6294, 1.6294]	0.8862	1.3333
bsgs-interleaved	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	1.1347	1.1347	1.1347	1.1347	0.0000	[1.1347, 1.1347]	0.8862	1.3333

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
bsgs-interleaved	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.1934	1.1934	1.1934	1.1934	0.0000	[1.1934, 1.1934]	0.8862	1.3333
bsgs-interleaved	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	0.5107	0.5107	0.5107	0.5107	0.0000	[0.5107, 0.5107]	0.8862	1.3333
bsgs-interleaved	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	0.6535	0.6535	0.6535	0.6535	0.0000	[0.6535, 0.6535]	0.8862	1.3333
bsgs-interleaved	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	1.8594	1.8594	1.8594	1.8594	0.0000	[1.8594, 1.8594]	0.8862	1.3333
bsgs-interleaved	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.3924	1.3924	1.3924	1.3924	0.0000	[1.3924, 1.3924]	0.8862	1.3333
bsgs-interleaved	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	1.8782	1.8782	1.8782	1.8782	0.0000	[1.8782, 1.8782]	0.8862	1.3333
bsgs-interleaved	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.5555	1.5555	1.5555	1.5555	0.0000	[1.5555, 1.5555]	0.8862	1.3333
bsgs-interleaved	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.5191	1.5191	1.5191	1.5191	0.0000	[1.5191, 1.5191]	0.8862	1.3333
bsgs-interleaved	W1062bb0f2e32	icv1-fp28-tm3161-380f655a	3 / 3	1.8637	1.8637	1.8637	1.8637	0.0000	[1.8637, 1.8637]	0.8862	1.3333
bsgs-interleaved	W08bb71dc3759	icv1-fp28-tm3161-380f655a	3 / 3	1.5380	1.5380	1.5380	1.5380	0.0000	[1.5380, 1.5380]	0.8862	1.3333
bsgs-interleaved	W0d13e64ff5c5	icv1-fp30-t31711-b403411a	3 / 3	1.3530	1.3530	1.3530	1.3530	0.0000	[1.3530, 1.3530]	0.8862	1.3333
bsgs-interleaved	W1670e908d864	icv1-fp30-t31711-b403411a	3 / 3	1.0437	1.0437	1.0437	1.0437	0.0000	[1.0437, 1.0437]	0.8862	1.3333
bsgs-interleaved	W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	3 / 3	1.9717	1.9717	1.9717	1.9717	0.0000	[1.9717, 1.9717]	0.8862	1.3333
bsgs-interleaved	W751e52089a7e	icv1-fp32-t74357-b7396dac	3 / 3	0.7059	0.7059	0.7059	0.7059	0.0000	[0.7059, 0.7059]	0.8862	1.3333
bsgs-negation	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	0.8765	0.8765	0.8765	0.8765	0.0000	[0.8765, 0.8765]	0.8862	1.0000
bsgs-negation	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	1.4058	1.4058	1.4058	1.4058	0.0000	[1.4058, 1.4058]	0.8862	1.0000
bsgs-negation	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	0.5554	0.5554	0.5554	0.5554	0.0000	[0.5554, 0.5554]	0.8862	1.0000
bsgs-negation	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	1.0826	1.0826	1.0826	1.0826	0.0000	[1.0826, 1.0826]	0.8862	1.0000
bsgs-negation	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.1081	1.1081	1.1081	1.1081	0.0000	[1.1081, 1.1081]	0.8862	1.0000
bsgs-negation	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	0.7667	0.7667	0.7667	0.7667	0.0000	[0.7667, 0.7667]	0.8862	1.0000
bsgs-negation	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	0.8323	0.8323	0.8323	0.8323	0.0000	[0.8323, 0.8323]	0.8862	1.0000
bsgs-negation	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	0.7227	0.7227	0.7227	0.7227	0.0000	[0.7227, 0.7227]	0.8862	1.0000
bsgs-negation	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.1988	1.1988	1.1988	1.1988	0.0000	[1.1988, 1.1988]	0.8862	1.0000
bsgs-negation	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	1.4414	1.4414	1.4414	1.4414	0.0000	[1.4414, 1.4414]	0.8862	1.0000
bsgs-negation	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	0.8677	0.8677	0.8677	0.8677	0.0000	[0.8677, 0.8677]	0.8862	1.0000
bsgs-negation	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	0.9761	0.9761	0.9761	0.9761	0.0000	[0.9761, 0.9761]	0.8862	1.0000
bsgs-negation	W1062bb0f2e32	icv1-fp28-tm3161-380f655a	3 / 3	0.7424	0.7424	0.7424	0.7424	0.0000	[0.7424, 0.7424]	0.8862	1.0000
bsgs-negation	W08bb71dc3759	icv1-fp28-tm3161-380f655a	3 / 3	0.5443	0.5443	0.5443	0.5443	0.0000	[0.5443, 0.5443]	0.8862	1.0000
bsgs-negation	W0d13e64ff5c5	icv1-fp30-t31711-b403411a	3 / 3	0.7873	0.7873	0.7873	0.7873	0.0000	[0.7873, 0.7873]	0.8862	1.0000
bsgs-negation	W1670e908d864	icv1-fp30-t31711-b403411a	3 / 3	1.0222	1.0222	1.0222	1.0222	0.0000	[1.0222, 1.0222]	0.8862	1.0000
bsgs-negation	W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	3 / 3	1.4860	1.4860	1.4860	1.4860	0.0000	[1.4860, 1.4860]	0.8862	1.0000
bsgs-negation	W751e52089a7e	icv1-fp32-t74357-b7396dac	3 / 3	0.8531	0.8531	0.8531	0.8531	0.0000	[0.8531, 0.8531]	0.8862	1.0000
kangaroo-vow	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	1.8744	1.8744	1.8744	1.8744	0.0000	[1.8744, 1.8744]	0.8862	2.0000
kangaroo-vow	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	1.6618	1.6618	1.6618	1.6618	0.0000	[1.6618, 1.6618]	0.8862	2.0000
kangaroo-vow	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	3.2978	3.2978	3.2978	3.2978	0.0000	[3.2978, 3.2978]	0.8862	2.0000
kangaroo-vow	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	0.5728	0.5728	0.5728	0.5728	0.0000	[0.5728, 0.5728]	0.8862	2.0000
kangaroo-vow	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	2.3762	2.3762	2.3762	2.3762	0.0000	[2.3762, 2.3762]	0.8862	2.0000
kangaroo-vow	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	1.5188	1.5188	1.5188	1.5188	0.0000	[1.5188, 1.5188]	0.8862	2.0000
kangaroo-vow	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	2.4265	2.4265	2.4265	2.4265	0.0000	[2.4265, 2.4265]	0.8862	2.0000
kangaroo-vow	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	2.9160	2.9160	2.9160	2.9160	0.0000	[2.9160, 2.9160]	0.8862	2.0000
kangaroo-vow	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	2.1296	2.1296	2.1296	2.1296	0.0000	[2.1296, 2.1296]	0.8862	2.0000
kangaroo-vow	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	3.1785	3.1785	3.1785	3.1785	0.0000	[3.1785, 3.1785]	0.8862	2.0000
kangaroo-vow	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.5013	1.5013	1.5013	1.5013	0.0000	[1.5013, 1.5013]	0.8862	2.0000
kangaroo-vow	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	0.6386	0.6386	0.6386	0.6386	0.0000	[0.6386, 0.6386]	0.8862	2.0000
kangaroo-vow	W1062bb0f2e32	icv1-fp28-tm3161-380f655a	3 / 3	2.7230	2.7230	2.7230	2.7230	0.0000	[2.7230, 2.7230]	0.8862	2.0000
kangaroo-vow	W08bb71dc3759	icv1-fp28-tm3161-380f655a	3 / 3	3.2746	3.2746	3.2746	3.2746	0.0000	[3.2746, 3.2746]	0.8862	2.0000

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
kangaroo-vow	W0d13e64ff5c5	icv1-fp30-t31711-b403411a	3 / 3	0.9103	0.9103	0.9103	0.9103	0.0000	[0.9103, 0.9103]	0.8862	2.0000
kangaroo-vow	W1670e908d864	icv1-fp30-t31711-b403411a	3 / 3	2.0528	2.0528	2.0528	2.0528	0.0000	[2.0528, 2.0528]	0.8862	2.0000
kangaroo-vow	W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	3 / 3	2.2029	2.2029	2.2029	2.2029	0.0000	[2.2029, 2.2029]	0.8862	2.0000
kangaroo-vow	W751e52089a7e	icv1-fp32-t74357-b7396dac	3 / 3	1.0300	1.0300	1.0300	1.0300	0.0000	[1.0300, 1.0300]	0.8862	2.0000
rho-negation-aa	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	1.7414	1.7399	1.6054	1.8788	0.1367	[1.6054, 1.8788]	0.8862	0.8862
rho-negation-aa	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	2.0812	2.2215	1.1021	2.9201	0.9171	[1.1021, 2.9201]	0.8862	0.8862
rho-negation-aa	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	1.3148	1.4146	1.0219	1.5079	0.2579	[1.0219, 1.5079]	0.8862	0.8862
rho-negation-aa	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	2.0554	1.6337	1.1586	3.3738	1.1662	[1.1586, 3.3738]	0.8862	0.8862
rho-negation-aa	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.4877	1.3548	1.0787	2.0295	0.4891	[1.0787, 2.0295]	0.8862	0.8862
rho-negation-aa	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	1.1494	0.8641	0.8081	1.7761	0.5435	[0.8081, 1.7761]	0.8862	0.8862
rho-negation-aa	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0978	1.0944	1.0508	1.1482	0.0488	[1.0508, 1.1482]	0.8862	0.8862
rho-negation-aa	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0937	0.8772	0.8048	1.5993	0.4393	[0.8048, 1.5993]	0.8862	0.8862
rho-negation-aa	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.3951	1.6850	0.3845	2.1159	0.9014	[0.3845, 2.1159]	0.8862	0.8862
rho-negation-aa	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	0.9218	0.9197	0.5495	1.2961	0.3733	[0.5495, 1.2961]	0.8862	0.8862
rho-negation-aa	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	0.5281	0.6728	0.0859	0.8258	0.3906	[0.0859, 0.8258]	0.8862	0.8862
rho-negation-aa	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	2.2182	2.1793	1.6953	2.7800	0.5434	[1.6953, 2.7800]	0.8862	0.8862
rho-negation-aa	W1062bb0f2e32	icv1-fp28-tm3161-380f655a	3 / 3	1.0970	1.3563	0.4423	1.4923	0.5711	[0.4423, 1.4923]	0.8862	0.8862
rho-negation-aa	W08bb71dc3759	icv1-fp28-tm3161-380f655a	3 / 3	1.0003	1.0215	0.7330	1.2463	0.2573	[0.7330, 1.2463]	0.8862	0.8862
rho-negation-aa	W0d13e64ff5c5	icv1-fp30-t31711-b403411a	3 / 3	1.1006	0.9256	0.3359	2.0404	0.8656	[0.3359, 2.0404]	0.8862	0.8862
rho-negation-aa	W1670e908d864	icv1-fp30-t31711-b403411a	3 / 3	1.0647	1.2862	0.4802	1.4275	0.5111	[0.4802, 1.4275]	0.8862	0.8862
rho-negation-aa	W1f9e5fbf5a84	icv1-fp32-t74357-b7396dac	3 / 3	0.9321	0.8435	0.4845	1.4684	0.4979	[0.4845, 1.4684]	0.8862	0.8862
rho-negation-aa	W751e52089a7e	icv1-fp32-t74357-b7396dac	3 / 3	0.8922	0.9494	0.4964	1.2309	0.3706	[0.4964, 1.2309]	0.8862	0.8862

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-plain	W43a0abf0d808	3 / 3	539.3	2.6406	26.7	366.7	1349.7	629.7	539.3	4.9625	41.1060	admitted (≥L2)
rho-plain	Wd0d7d5ab0c1f	3 / 3	413.7	2.0253	12.3	245.0	1101.3	505.0	413.7	5.4433	36.0140	admitted (≥L2)
rho-plain	Wbfca0f4fc222	3 / 3	743.0	1.8190	42.3	547.0	1762.0	835.0	743.0	5.5068	36.4660	admitted (≥L2)
rho-plain	Wd5517e0db0bc	3 / 3	586.0	1.4346	24.3	398.7	1460.0	682.0	586.0	5.5373	34.8460	informative, L1
rho-plain	W141767324d6e	3 / 3	1364.3	2.0528	27.7	1170.0	3025.7	1463.3	1364.3	5.8921	34.8060	admitted (≥L2)
rho-plain	W8d7a772f8ccd	3 / 3	797.0	1.1992	24.7	611.3	1878.0	891.7	797.0	5.4149	31.4820	informative, L1
rho-plain	W9a300e433cac	3 / 3	1587.3	1.1476	42.0	1370.3	3483.7	1690.3	1587.3	6.0073	37.0510	admitted (≥L2)
rho-plain	W6e13d2547daa	3 / 3	2888.0	2.0880	74.7	2643.0	6073.0	2987.0	2888.0	6.3761	39.2720	admitted (≥L2)
rho-plain	Wec44ca6ca670	3 / 3	6332.3	1.7600	97.3	5864.7	13309.7	6547.3	6332.3	7.3040	34.2070	informative, L1
rho-plain	W3256bf1f8f74	3 / 3	3603.0	1.0014	49.3	3181.7	7858.0	3820.3	3603.0	8.1698	38.5700	informative, L1
rho-plain	Wf9c5da9613ce	3 / 3	9746.0	1.6838	143.0	9227.0	20151.0	9965.7	9746.0	8.0300	36.1810	admitted (≥L2)
rho-plain	Wbee2c4f33e98	3 / 3	7622.3	1.3169	106.7	7146.7	15902.7	7841.7	7622.3	6.9392	33.4880	informative, L1
rho-plain	W1062bb0f2e32	3 / 3	13865.7	0.9699	98.7	13381.7	28401.3	14089.0	13865.7	6.9102	37.4160	admitted (≥L2)
rho-plain	W08bb71dc3759	3 / 3	22442.7	1.5699	162.3	21898.7	45549.3	22664.0	22442.7	8.0620	44.3990	admitted (≥L2)
rho-plain	W0d13e64ff5c5	3 / 3	37521.0	1.3602	284.7	36845.7	75723.0	37748.0	37521.0	10.2159	40.9410	informative, L1
rho-plain	W1670e908d864	3 / 3	54647.7	1.9811	408.0	53857.3	109975.3	54874.3	54647.7	12.9273	40.8050	admitted (≥L2)
rho-plain	W1f9e5fbf5a84	3 / 3	65931.3	1.2636	265.3	65265.0	132562.7	66164.7	65931.3	16.6386	43.6560	informative, L1
rho-plain	W751e52089a7e	3 / 3	92600.0	1.7746	363.3	91845.0	185898.0	92832.7	92600.0	19.2823	51.4320	informative, L1
rho-negation	W43a0abf0d808	3 / 3	401.3	1.9649	14.3	225.7	1075.7	492.3	401.3	5.2955	42.1090	admitted (≥L2)
rho-negation	Wd0d7d5ab0c1f	3 / 3	479.7	2.3484	15.7	283.3	1246.3	575.3	479.7	5.2413	33.1740	informative, L1
rho-negation	Wbfca0f4fc222	3 / 3	606.0	1.4836	23.3	400.7	1494.0	700.0	606.0	5.7177	32.7820	informative, L1
rho-negation	Wd5517e0db0bc	3 / 3	947.3	2.3192	44.3	690.0	2196.7	1048.0	947.3	5.8911	35.4580	informative, L1

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-negation	W141767324d6e	3 / 3	1115.7	1.6786	21.0	862.0	2536.3	1217.3	1115.7	5.5466	33.3640	admitted (≥L2)
rho-negation	W8d7a772f8ccd	3 / 3	862.0	1.2970	18.0	640.0	2015.0	959.0	862.0	5.3495	33.3050	informative, L1
rho-negation	W9a300e433cac	3 / 3	1713.3	1.2387	46.0	1392.7	3761.7	1825.0	1713.3	7.0971	43.6870	informative, L1
rho-negation	W6e13d2547daa	3 / 3	1707.0	1.2341	42.7	1375.3	3738.0	1815.0	1707.0	5.9879	33.4990	admitted (≥L2)
rho-negation	Wec44ca6ca670	3 / 3	5664.0	1.5742	78.0	5036.7	11991.0	5885.0	5664.0	6.7924	35.3500	informative, L1
rho-negation	W3256bf1f8f74	3 / 3	3742.3	1.0401	43.3	3218.0	8141.7	3961.3	3742.3	7.9113	39.9310	informative, L1
rho-negation	Wf9c5da9613ce	3 / 3	3449.3	0.5960	48.3	2925.3	7569.7	3673.0	3449.3	6.6998	34.1670	admitted (≥L2)
rho-negation	Wbee2c4f33e98	3 / 3	14487.0	2.5030	202.0	13462.3	29680.0	14722.3	14487.0	8.2803	34.8180	admitted (≥L2)
rho-negation	W1062bb0f2e32	3 / 3	17694.7	1.2378	132.3	16605.3	36125.3	17940.0	17694.7	9.7309	45.2950	admitted (≥L2)
rho-negation	W08bb71dc3759	3 / 3	16135.3	1.1287	115.3	15128.7	32991.7	16375.7	16135.3	8.9867	40.5320	informative, L1
rho-negation	W0d13e64ff5c5	3 / 3	34258.0	1.2419	260.7	32482.7	69312.0	34523.3	34258.0	12.9904	47.5550	informative, L1
rho-negation	W1670e908d864	3 / 3	33138.3	1.2014	236.3	31456.0	67060.7	33399.7	33138.3	10.8797	40.2590	admitted (≥L2)
rho-negation	W1f9e5fbf5a84	3 / 3	54881.3	1.0518	198.3	52491.7	110648.7	55176.7	54881.3	13.6365	42.8790	admitted (≥L2)
rho-negation	W751e52089a7e	3 / 3	52532.0	1.0068	182.7	50247.7	105940.0	52824.0	52532.0	16.4684	46.6020	informative, L1
bsgs-textbook	W43a0abf0d808	3 / 3	317.0	1.5520	–	306.0	656.0	324.0	316.0	4.9998	36.7710	informative, L1
bsgs-textbook	Wd0d7d5ab0c1f	3 / 3	441.0	2.1591	–	430.0	904.0	448.0	440.0	5.2699	35.3570	informative, L1
bsgs-textbook	Wbfca0f4fc222	3 / 3	483.0	1.1825	–	471.0	991.0	491.0	482.0	5.4962	34.1630	admitted (≥L2)
bsgs-textbook	Wd5517e0db0bc	3 / 3	728.0	1.7823	–	716.0	1481.0	736.0	727.0	6.9678	36.1710	informative, L1
bsgs-textbook	W141767324d6e	3 / 3	1204.0	1.8115	–	1190.0	2436.0	1213.0	1203.0	5.2299	35.9720	informative, L1
bsgs-textbook	W8d7a772f8ccd	3 / 3	948.0	1.4264	–	934.0	1924.0	957.0	947.0	5.6978	32.2320	informative, L1
bsgs-textbook	W9a300e433cac	3 / 3	2077.0	1.5017	–	2064.0	4185.0	2087.0	2076.0	6.8705	44.9040	admitted (≥L2)
bsgs-textbook	W6e13d2547daa	3 / 3	1906.0	1.3780	–	1893.0	3843.0	1916.0	1905.0	5.3673	40.6450	informative, L1
bsgs-textbook	Wec44ca6ca670	3 / 3	6895.0	1.9164	–	6877.0	13824.0	6906.0	6894.0	5.2730	29.6930	admitted (≥L2)
bsgs-textbook	W3256bf1f8f74	3 / 3	7881.0	2.1904	–	7863.0	15796.0	7892.0	7880.0	5.9269	35.2750	admitted (≥L2)
bsgs-textbook	Wf9c5da9613ce	3 / 3	8931.0	1.5430	–	8915.0	17899.0	8943.0	8930.0	6.8294	39.3670	informative, L1
bsgs-textbook	Wbee2c4f33e98	3 / 3	9639.0	1.6654	–	9623.0	19315.0	9651.0	9638.0	6.2909	33.4560	informative, L1
bsgs-textbook	W1062bb0f2e32	3 / 3	20040.0	1.4019	–	20021.0	40120.0	20053.0	20039.0	10.5521	50.2200	admitted (≥L2)
bsgs-textbook	W08bb71dc3759	3 / 3	16843.0	1.1782	–	16824.0	33726.0	16856.0	16842.0	7.1873	38.3030	admitted (≥L2)
bsgs-textbook	W0d13e64ff5c5	3 / 3	40067.0	1.4525	–	40046.0	80177.0	40081.0	40066.0	13.3667	45.6200	admitted (≥L2)
bsgs-textbook	W1670e908d864	3 / 3	47379.0	1.7176	–	47358.0	94801.0	47393.0	47378.0	12.7865	40.3610	admitted (≥L2)
bsgs-textbook	W1f9e5fbf5a84	3 / 3	116937.0	2.2411	–	116911.0	233920.0	116952.0	116936.0	27.7786	52.1870	informative, L1
bsgs-textbook	W751e52089a7e	3 / 3	79672.0	1.5269	–	79646.0	159390.0	79687.0	79671.0	23.5444	57.6850	informative, L1
bsgs-interleaved	W43a0abf0d808	3 / 3	438.0	2.1444	–	426.0	898.0	445.0	437.0	5.8273	42.6940	informative, L1
bsgs-interleaved	Wd0d7d5ab0c1f	3 / 3	410.0	2.0073	–	398.0	842.0	417.0	409.0	5.2680	40.6620	admitted (≥L2)
bsgs-interleaved	Wbfca0f4fc222	3 / 3	751.0	1.8386	–	738.0	1527.0	759.0	750.0	5.4200	33.2160	admitted (≥L2)
bsgs-interleaved	Wd5517e0db0bc	3 / 3	523.0	1.2804	–	510.0	1071.0	531.0	522.0	6.6674	36.8540	admitted (≥L2)
bsgs-interleaved	W141767324d6e	3 / 3	895.0	1.3466	–	880.0	1818.0	904.0	894.0	5.2940	32.2700	informative, L1
bsgs-interleaved	W8d7a772f8ccd	3 / 3	383.0	0.5763	–	368.0	794.0	392.0	382.0	5.1579	35.9400	informative, L1
bsgs-interleaved	W9a300e433cac	3 / 3	1020.0	0.7375	–	1006.0	2071.0	1030.0	1019.0	7.5814	45.5760	informative, L1
bsgs-interleaved	W6e13d2547daa	3 / 3	2902.0	2.0981	–	2888.0	5835.0	2912.0	2901.0	6.6138	42.1630	informative, L1
bsgs-interleaved	Wec44ca6ca670	3 / 3	5653.0	1.5712	–	5634.0	11340.0	5664.0	5652.0	7.2920	36.8820	informative, L1
bsgs-interleaved	W3256bf1f8f74	3 / 3	7625.0	2.1193	–	7606.0	15284.0	7636.0	7624.0	8.4274	39.4170	admitted (≥L2)
bsgs-interleaved	Wf9c5da9613ce	3 / 3	10159.0	1.7552	–	10142.0	20355.0	10171.0	10158.0	7.7724	36.2960	informative, L1
bsgs-interleaved	Wbee2c4f33e98	3 / 3	9921.0	1.7141	–	9904.0	19879.0	9933.0	9920.0	7.6186	33.6940	admitted (≥L2)
bsgs-interleaved	W1062bb0f2e32	3 / 3	30062.0	2.1029	–	30042.0	60164.0	30075.0	30061.0	14.7197	43.0720	admitted (≥L2)
bsgs-interleaved	W08bb71dc3759	3 / 3	24808.0	1.7354	–	24788.0	49656.0	24821.0	24807.0	11.9890	39.2690	informative, L1

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
bsgs-interleaved	W0d13e64ff5c5	3 / 3	42114.0	1.5267	–	42092.0	84271.0	42128.0	42113.0	17.0475	51.2630	informative, L1
bsgs-interleaved	W1670e908d864	3 / 3	32486.0	1.1777	–	32464.0	65015.0	32500.0	32485.0	12.8209	41.6730	admitted (≥L2)
bsgs-interleaved	W1f9e5fbf5a84	3 / 3	116091.0	2.2248	–	116064.0	232228.0	116106.0	116090.0	35.2299	67.1630	informative, L1
bsgs-interleaved	W751e52089a7e	3 / 3	41561.0	0.7965	–	41534.0	83168.0	41576.0	41560.0	19.8268	64.6770	admitted (≥L2)
bsgs-negation	W43a0abf0d808	3 / 3	202.0	0.9890	–	192.0	426.0	209.0	201.0	5.9019	38.4560	admitted (≥L2)
bsgs-negation	Wd0d7d5ab0c1f	3 / 3	324.0	1.5863	–	314.0	670.0	331.0	323.0	4.9982	36.4070	informative, L1
bsgs-negation	Wbfca0f4fc222	3 / 3	256.0	0.6267	–	243.0	537.0	264.0	255.0	5.0102	32.3390	informative, L1
bsgs-negation	Wd5517e0db0bc	3 / 3	499.0	1.2216	–	486.0	1023.0	507.0	498.0	6.3210	40.3810	informative, L1
bsgs-negation	W141767324d6e	3 / 3	831.0	1.2503	–	816.0	1690.0	840.0	830.0	4.8224	32.0440	informative, L1
bsgs-negation	W8d7a772f8ccd	3 / 3	575.0	0.8651	–	560.0	1178.0	584.0	574.0	4.9697	32.5000	informative, L1
bsgs-negation	W9a300e433cac	3 / 3	1299.0	0.9392	–	1285.0	2629.0	1309.0	1298.0	5.9769	44.6590	admitted (≥L2)
bsgs-negation	W6e13d2547daa	3 / 3	1128.0	0.8155	–	1114.0	2287.0	1138.0	1127.0	4.8612	31.9740	admitted (≥L2)
bsgs-negation	Wec44ca6ca670	3 / 3	4867.0	1.3527	–	4848.0	9768.0	4878.0	4866.0	5.6784	32.9880	admitted (≥L2)
bsgs-negation	W3256bf1f8f74	3 / 3	5852.0	1.6265	–	5833.0	11738.0	5863.0	5851.0	6.0755	35.0550	informative, L1
bsgs-negation	Wf9c5da9613ce	3 / 3	5667.0	0.9791	–	5651.0	11371.0	5679.0	5666.0	5.9927	32.8700	informative, L1
bsgs-negation	Wbee2c4f33e98	3 / 3	6375.0	1.1014	–	6359.0	12787.0	6387.0	6374.0	5.9155	36.5260	admitted (≥L2)
bsgs-negation	W1062bb0f2e32	3 / 3	11976.0	0.8378	–	11957.0	23992.0	11989.0	11975.0	8.0881	43.2160	informative, L1
bsgs-negation	W08bb71dc3759	3 / 3	8780.0	0.6142	–	8761.0	17600.0	8793.0	8779.0	6.7520	36.8920	admitted (≥L2)
bsgs-negation	W0d13e64ff5c5	3 / 3	24506.0	0.8884	–	24484.0	49055.0	24520.0	24505.0	9.8102	41.0070	admitted (≥L2)
bsgs-negation	W1670e908d864	3 / 3	31818.0	1.1535	–	31796.0	63679.0	31832.0	31817.0	10.6927	40.0380	informative, L1
bsgs-negation	W1f9e5fbf5a84	3 / 3	87490.0	1.6767	–	87471.0	175026.0	87505.0	87489.0	18.9860	48.1220	admitted (≥L2)
bsgs-negation	W751e52089a7e	3 / 3	50227.0	0.9626	–	50208.0	100500.0	50242.0	50226.0	15.9443	57.5070	admitted (≥L2)
kangaroo-vow	W43a0abf0d808	3 / 3	432.0	2.1151	52.0	398.0	931.0	454.0	431.0	5.4052	42.4580	informative, L1
kangaroo-vow	Wd0d7d5ab0c1f	3 / 3	383.0	1.8752	50.0	349.0	833.0	405.0	382.0	5.1418	32.7550	informative, L1
kangaroo-vow	Wbfca0f4fc222	3 / 3	1520.0	3.7212	191.0	1483.0	3116.0	1545.0	1519.0	5.3931	32.5870	informative, L1
kangaroo-vow	Wd5517e0db0bc	3 / 3	264.0	0.6463	33.0	227.0	604.0	289.0	263.0	5.3784	33.1310	informative, L1
kangaroo-vow	W141767324d6e	3 / 3	1782.0	2.6812	105.0	1743.0	3649.0	1810.0	1781.0	4.9410	32.2020	informative, L1
kangaroo-vow	W8d7a772f8ccd	3 / 3	1139.0	1.7137	70.0	1100.0	2363.0	1167.0	1138.0	5.2729	33.1240	informative, L1
kangaroo-vow	W9a300e433cac	3 / 3	3787.0	2.7380	217.0	3746.0	7668.0	3818.0	3786.0	6.8884	49.7590	informative, L1
kangaroo-vow	W6e13d2547daa	3 / 3	4551.0	3.2903	278.0	4510.0	9196.0	4582.0	4550.0	6.6191	36.8150	informative, L1
kangaroo-vow	Wec44ca6ca670	3 / 3	8646.0	2.4030	255.0	8593.0	17395.0	8680.0	8645.0	6.4953	31.2530	informative, L1
kangaroo-vow	W3256bf1f8f74	3 / 3	12904.0	3.5865	400.0	12851.0	25911.0	12938.0	12903.0	7.2778	34.7010	informative, L1
kangaroo-vow	Wf9c5da9613ce	3 / 3	9805.0	1.6940	310.0	9754.0	19722.0	9842.0	9804.0	6.6530	34.1070	admitted (≥L2)
kangaroo-vow	Wbee2c4f33e98	3 / 3	4171.0	0.7206	129.0	4120.0	8454.0	4208.0	4170.0	6.3674	37.8600	admitted (≥L2)
kangaroo-vow	W1062bb0f2e32	3 / 3	43924.0	3.0726	697.0	43866.0	87972.0	43965.0	43923.0	11.0130	43.3710	admitted (≥L2)
kangaroo-vow	W08bb71dc3759	3 / 3	52821.0	3.6950	844.0	52763.0	105766.0	52862.0	52820.0	12.0527	39.9260	admitted (≥L2)
kangaroo-vow	W0d13e64ff5c5	3 / 3	28332.0	1.0271	443.0	28270.0	56797.0	28376.0	28331.0	8.6515	38.0880	admitted (≥L2)
kangaroo-vow	W1670e908d864	3 / 3	63893.0	2.3163	970.0	63831.0	127919.0	63937.0	63892.0	12.7725	42.9930	informative, L1
kangaroo-vow	W1f9e5fbf5a84	3 / 3	129701.0	2.4857	1012.0	129640.0	259544.0	129748.0	129700.0	21.8958	48.2450	informative, L1
kangaroo-vow	W751e52089a7e	3 / 3	60647.0	1.1623	472.0	60586.0	121436.0	60694.0	60646.0	13.4638	45.4630	admitted (≥L2)
rho-negation-aa	W43a0abf0d808	3 / 3	401.3	1.9649	14.3	225.7	1075.7	492.3	401.3	5.3545	43.9740	informative, L1
rho-negation-aa	Wd0d7d5ab0c1f	3 / 3	479.7	2.3484	15.7	283.3	1246.3	575.3	479.7	5.0281	35.4290	informative, L1
rho-negation-aa	Wbfca0f4fc222	3 / 3	606.0	1.4836	23.3	400.7	1494.0	700.0	606.0	5.3045	31.5060	informative, L1
rho-negation-aa	Wd5517e0db0bc	3 / 3	947.3	2.3192	44.3	690.0	2196.7	1048.0	947.3	5.7143	32.8540	informative, L1
rho-negation-aa	W141767324d6e	3 / 3	1115.7	1.6786	21.0	862.0	2536.3	1217.3	1115.7	5.4187	34.3350	admitted (≥L2)
rho-negation-aa	W8d7a772f8ccd	3 / 3	862.0	1.2970	18.0	640.0	2015.0	959.0	862.0	5.6938	34.6280	admitted (≥L2)

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-negation-aa	W9a300e433cac	3 / 3	1713.3	1.2387	46.0	1392.7	3761.7	1825.0	1713.3	7.1680	40.7020	informative, L1
rho-negation-aa	W6e13d2547daa	3 / 3	1707.0	1.2341	42.7	1375.3	3738.0	1815.0	1707.0	6.4226	39.9230	admitted (≥L2)
rho-negation-aa	Wec44ca6ca670	3 / 3	5664.0	1.5742	78.0	5036.7	11991.0	5885.0	5664.0	6.8737	35.0820	informative, L1
rho-negation-aa	W3256bf1f8f74	3 / 3	3742.3	1.0401	43.3	3218.0	8141.7	3961.3	3742.3	7.2805	35.3720	admitted (≥L2)
rho-negation-aa	Wf9c5da9613ce	3 / 3	3449.3	0.5960	48.3	2925.3	7569.7	3673.0	3449.3	5.7817	32.1710	admitted (≥L2)
rho-negation-aa	Wbee2c4f33e98	3 / 3	14487.0	2.5030	202.0	13462.3	29680.0	14722.3	14487.0	7.7729	39.8280	admitted (≥L2)
rho-negation-aa	W1062bb0f2e32	3 / 3	17694.7	1.2378	132.3	16605.3	36125.3	17940.0	17694.7	10.4982	45.6310	admitted (≥L2)
rho-negation-aa	W08bb71dc3759	3 / 3	16135.3	1.1287	115.3	15128.7	32991.7	16375.7	16135.3	7.4309	32.3410	informative, L1
rho-negation-aa	W0d13e64ff5c5	3 / 3	34258.0	1.2419	260.7	32482.7	69312.0	34523.3	34258.0	10.0419	39.7770	admitted (≥L2)
rho-negation-aa	W1670e908d864	3 / 3	33138.3	1.2014	236.3	31456.0	67060.7	33399.7	33138.3	10.7863	54.2410	informative, L1
rho-negation-aa	W1f9e5fbf5a84	3 / 3	54881.3	1.0518	198.3	52491.7	110648.7	55176.7	54881.3	14.3206	44.4780	informative, L1
rho-negation-aa	W751e52089a7e	3 / 3	52532.0	1.0068	182.7	50247.7	105940.0	52824.0	52532.0	15.6846	49.9640	admitted (≥L2)

Scaling

arm	sizes	verified runs	α [d5]	se(α)	log2 C	R^2
rho-plain	9	54	0.474	0.013	1.112	0.9946
rho-negation	9	54	0.447	0.012	1.570	0.9949
bsgs-textbook	9	54	0.499	0.014	0.545	0.9943
bsgs-interleaved	9	54	0.501	0.023	0.449	0.9860
bsgs-negation	9	54	0.498	0.022	−0.053	0.9867
kangaroo-vow	9	54	0.487	0.032	1.247	0.9714
rho-negation-aa	9	54	0.447	0.012	1.570	0.9949

Raw records

`research\/ecbench\_comprehensive\_sweep\_20261018\/sessions\/prime/records.jsonl` (504 records; ids in `raw_records.csv` and `report.json`); child stderr in `research\/ecbench\_comprehensive\_sweep\_20261018\/sessions\/prime/exec/` (0 file(s)).

Derivations

- **[d1]** S = cost.s of each verified, non-warm-up record (`total_gae / √r`, stored by the runner); mean, median, min, max and sample stdev (stats.rs) are over those records.
- **[d2]** 95% percentile bootstrap interval of the mean S: 4000 resamples of the verified runs of the cell with replacement (stats::bootstrap_ci, seed 20261002). A cell with one run has no interval. Not a significance test.
- **[d3]** total gae and S / floor are means over the verified runs of the cell of cost.total_gae and boundaries.ratio_to_floor; floor S is boundaries.floor_s = √(π / 2A), A = automorphisms the curve offers a generic method.
- **[d4]** expected S is methods::expected_s(method, A): the closed form the method's mean S approaches as r grows (set-up omitted). A reference mark, not a measurement.
- **[d5]** scaling: per arm, per curve, the mean gae over the arm's verified runs on that curve; the line is ordinary least squares of log2(mean gae) on log2(r) over the curves (bounds::ols), slope α, intercept log2 C so gae = C·r^α. The standard error of α is √(SS_res/(sizes−2)/S_xx) from those residuals (none below 3 sizes). Counted operations, one point per size: it is not the cluster-bootstrap interval of `ecbench bound fit`, which resamples runs and sizes.
- **[d6]** counters (distinguished points, steps) and field operations (muls, sqrs, invs) are means over the cell's verified runs of the counters the record carries (steps = `steps`, or baby + giant steps for BSGS); a dash means the method did not count it (unknown, not zero).
- **[d7]** wall and CPU are medians over the cell's verified runs of time.solve_wall_ns and time.user_ns + sys_ns. The figure is admitted only when every run behind it earned the level the spec requires (compare.rs's gate); otherwise it is informative at the lowest level earned, and never a result.
- **[d8]** phase share = the phase's mean gae over the sum of the arm's mean gae across phases on that curve, over verified runs.